

PATENT SPECIFICATION

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**TITLE: THE ADRIEL SHAMER WILLIS: TSVM
DEFENSE INFRASTRUCTURE AND
CONTROL LOOP ARCHITECTURE**

**TECHNICAL FIELD: QUANTUM-RESISTANT
CYBER-PHYSICAL AIR DEFENSE,
COGNITIVE ELECTRONIC WARFARE (CEW)
MITIGATION, AND DECENTRALIZED MULTI-
AGENT AUTONOMOUS MESH SYSTEMS**

ABSTRACT

A method and physical system architecture are disclosed for mitigating high-entropy cognitive electronic warfare (CEW) jamming and directed-energy weapons in autonomous drone arrays and multi-tiered counter-swarm air defense installations. The system rejects standard algorithmic software-based filtering in favor of a physical, hardware-enforced Information-Geometric Statistical Manifold control loop. By utilizing an automated, silicon-photonics-driven "Etch A Sketch" response-gain

modulation gate, the infrastructure strips out induced electrical cross-talk ("pressure phosphenes"). Concurrently, a Bayesian Rao-Blackwellized "Claw Machine" processing layer isolates low-dimensional sufficient statistics (Absolute Angle-of-Arrival and spatial geometric tensors) from chaotic wave fields. These invariants are routed via sub-threshold pre-biased memristor crossbars to trigger pre-simulated offline trajectory matrices ("Déjà Rêvé Plays") at a physical speed of less than 500 picoseconds. The architecture functions entirely offline, operating via multi-static earth-field geomagnetic tensor tracking and non-fuel kinetic/ RF energy-harvesting vacuum micro-diodes to render tactical airspaces invulnerable to saturation or interception.

THE CLAIMS

CLAIM 1: THE ADRIEL SHAMER WILLIS: TSVM PHYSICAL CONTROL SYSTEM

An un-jammable, hardware-enforced cyber-physical control loop architecture for decentralized multi-agent autonomous drone arrays and defensive counter-mesh nodes, the system comprising:

- **An Unfiltered Ingestion Layer (HSAM Substrate)**, characterized by the absolute structural decoupling of standard software-driven salience networks, configured to stream 100% of raw multi-modal sensor bitstreams (radio frequencies, phase variations, and telemetry fluctuations) directly to non-volatile, high-density Asymmetric Solid-State Flash Arrays;
- **An Inline Silicon Photonics Response-Gain Modulator ("Etch A Sketch" Noise-Gate)**, configured to physically drop input sensitivity across localized data buses to zero when exposed to an input electrical overvoltage surge exceeding a 40 Hz threshold, thereby phase-canceling induced electromagnetic harmonic ringing and cross-talk artifacts ("pressure phosphenes");
- **A Bayesian Rao-Blackwellized Estimation Engine ("Claw Machine" Filtration Layer)**, designed to extract a low-dimensional subset of sufficient statistics—specifically absolute wave Angle-of-Arrival (AoA) and spatial geometric tensors—from

high-entropy Shannon noise profiles while discarding non-invariant data payloads;

- **A Neuromorphic Memristor Crossbar Array Group**, pre-loaded with an offline-generated, generative LLM-simulated matrix of tactical counter-trajectories ("The Dream Vault"), wherein each discrete memristor junction is continuously maintained at a sub-threshold volatile voltage bias between 0.8V and 0.85V.

CLAIM 2: THE "ETCH A SKETCH" HIGH-FREQUENCY HARDWARE-CLEARING BUS

The system of Claim 1, wherein the inline Silicon Photonics Response-Gain Modulator utilizes a physical high-pass Gallium Nitride (GaN) field-effect transistor network that blocks low-frequency navigational loops while routing high-frequency (>40 Hz) cognitive electronic warfare jamming surges directly into an array of inhibitory optoelectronic switches, establishing instantaneous mechanical signal invariance via wave-field superposition.

CLAIM 3: THE "CLAW MACHINE" RAO-BLACKWELLIZED GEOMETRIC MATRIX GRIPPER

The system of Claim 1, wherein the Bayesian Rao-Blackwellized Estimation Engine implements a hardware-level statistical manifold filter that maps multi-static wave fronts mathematically to a Fisher Information Metric surface, converting raw analog voltage inputs into a single, rigid coordinate vector without digital Analog-to-Digital (ADC) preprocessing.

CLAIM 4: THE "DEJÀ RVÉ" ZERO-LATENCY TACTICAL BYPASS LOOP

The system of Claim 1, wherein the sub-threshold pre-biased memristor crossbar array utilizes the incoming analog voltage surge captured by Claim 3 to bridge a remaining 0.15V activation gap, inducing an immediate physical threshold switch ("silicon snap-through") in less than 500 picoseconds, shifting flight or interception control paths directly to a pre-computed tactical maneuver while completely bypassing the primary digital operating system and CPU code compilation stacks.

CLAIM 5: THE ZERO-FUEL STRUCTURAL ENERGETIC COUPLING GATE

An unshielded asymmetric wave-ingestion envelope integrated into the physical fuselage of the multi-agent array of Claim 1, comprising a multi-layered, fractal metamaterial outer shell coupled to an array of silicon-etched Field-Emission Vacuum Micro-Diodes, configured to capture wideband adversarial electromagnetic jamming fields and convert the induced high-frequency alternating currents into an isolated direct current (DC) voltage rail routed to a Graphene Ultracapacitor Group, thereby powering offline sub-threshold bias lines entirely via the adversary's offensive energy emission field.

CLAIM 6: THE MULTI-HEMISPHERE AMBIDEXTROUS MOTHERBOARD DUALISM

A symmetric hardware processing topology for the system of Claim 1, wherein the drone array or air defense counter-mesh node possesses twin, non-lateralized mirror microprocessors running parallel simulated Entorhinal-Hippocampal-Prefrontal control loops concurrently, ensuring that an electromagnetic or kinetic failure of the primary hemisphere initiates an instantaneous failover transition, granting the secondary hemisphere 100% operational command with zero-step context-switching latency.

CLAIM 7: THE GEOGRAPHIC ATTRACTOR OFFLINE NAVIGATION METHOD

The method of navigating a decentralized multi-agent array of Claim 1 under total optical, LiDAR, and GPS blindness conditions, the method comprising the continuous extraction of large-scale geological invariant tensors—specifically the Earth's natural geomagnetic fields and local micro-barometric low-frequency waves—to maintain an inline 3D Continuous Attractor Network (CAN) topology entirely isolated from external radio frequency manipulation.

CLAIM 8: THE MULTI-AGENT SYMPATHETIC CONSENSUS IMMUNE

SYSTEM

The decentralized multi-agent network of Claim 1, wherein the activation of a single node's sub-threshold threshold switch triggers a localized "flashbulb" snapshot of the attack vector within an air-gapped Hardware Security Module (HSM), while simultaneously broadcasting a post-quantum lattice-based digital signature (ML-DSA) hash to adjacent mesh entities, forcing neighboring nodes to dynamically scale their local volatile biases to 0.95V to execute a synchronized, sympathetic array-wide tactical counter-maneuver.